

درس ابزار دقیق



حسگرهای نوری و تابش حرارتی

مدرس: نوید علایی شینی

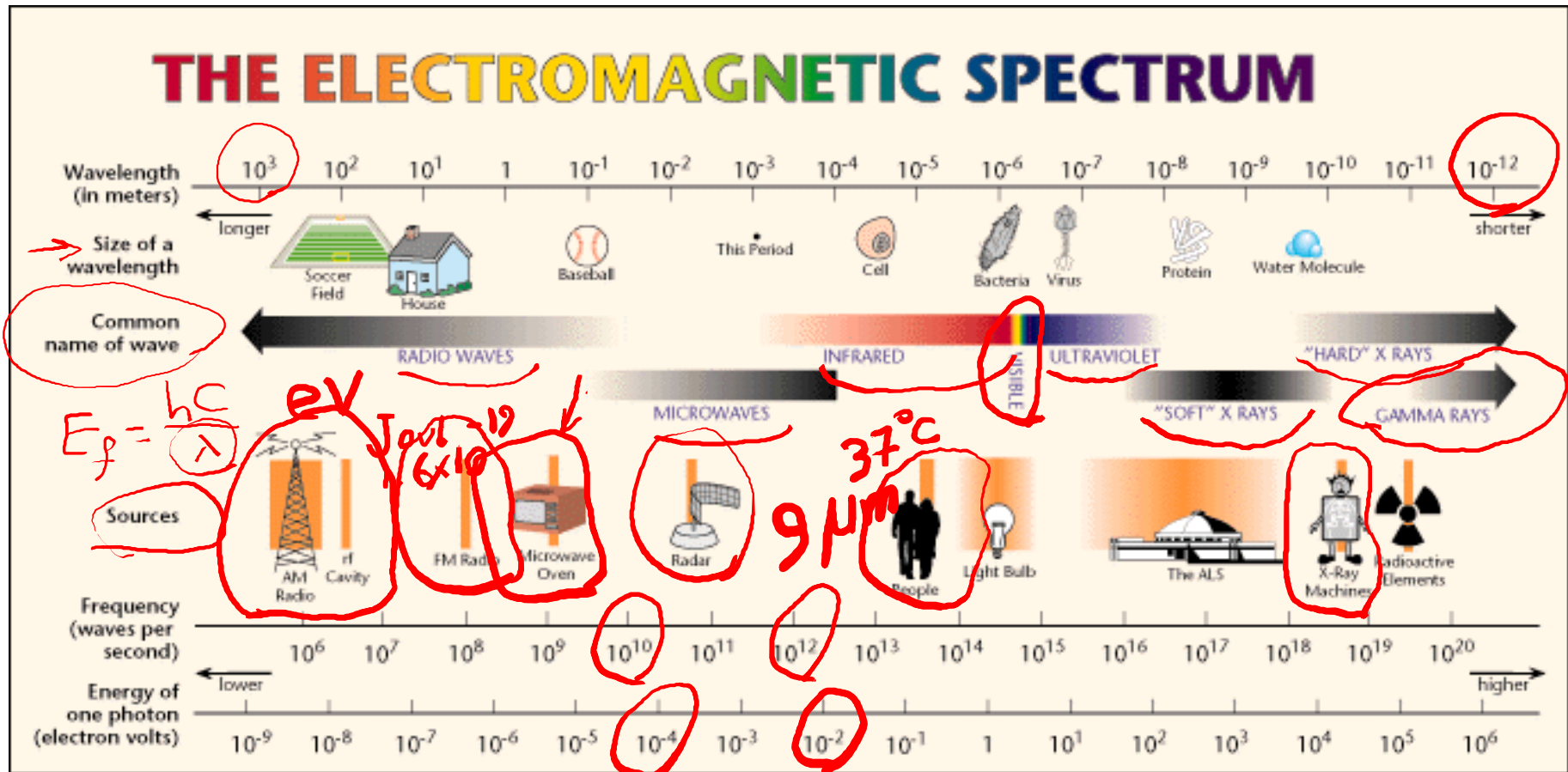
اسفندماه ۱۴۰۴ خورشیدی

فهرست:

- معرفی طیف الکترومغناطیسی
- قوانین تابش جسم سیاه (انتشار نور توسط اجسام گرم)
- واحدهای تابش نور
- دسته بندی آثار فوتونی
- اثر فتوالکتریک و مفهوم انرژی فوتون
- آشکارسازهای فتوکانداکتیو، میکروبالومتر
- تصویربردارها و حسگرهای حرارتی صنعتی
- آشکارسازهای فتوولتائیک
- آشکارسازهای فتودیودی PIN
- جذب نور در نیم رساناها
- کمیتهای مقایسه ای حسگرهای نوری

طیف الکترومغناطیسی

(ابعاد طول موج، نامگذاری، منابع تولید، محدوده ی فرکانس و انرژی)



طيف الكتر ومغناطيسي



Name	Wavelength	Frequency (Hz)	Photon Energy (eV)
Gamma ray	less than 0.01 nm	more than 30 EHz	124 keV – 300+ GeV
X-ray	0.01 nm – 10 nm	30 EHz – 30 PHz	124 eV – 124 keV
Ultraviolet	10 nm – 400 nm	30 PHz – 790 THz	3.3 eV – 124 eV
Visible	400 nm–700 nm	790 THz – 430 THz	1.7 eV – 3.3 eV
Infrared	700 nm – 1 mm	430 THz – 300 GHz	1.24 meV – 1.7 eV
Microwave	1 mm – 1 meter	300 GHz – 300 MHz	1.24 μeV – 1.24 meV
Radio	1 meter – 100,000 km	300 MHz – 3 Hz	12.4 feV – 1.24 μeV



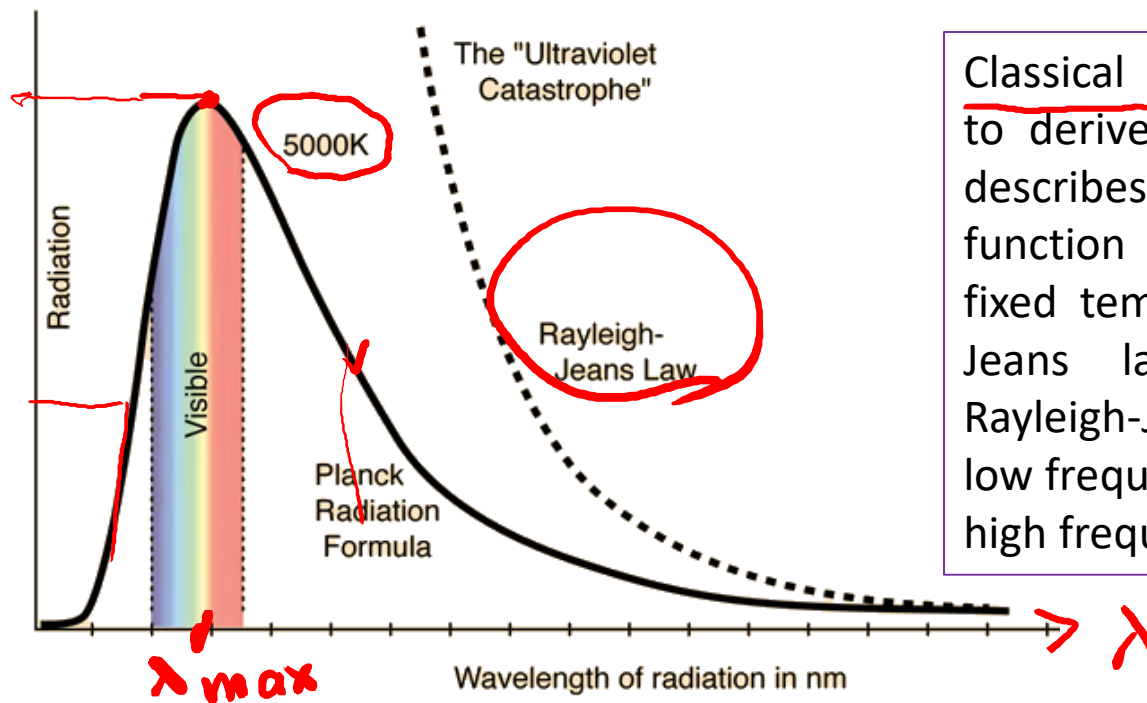
Color	Wavelength	Frequency	Photon energy
Violet	380–450 nm	668–789 THz	2.75–3.26 eV
Blue	450–495 nm	606–668 THz	2.50–2.75 eV
Green	495–570 nm	526–606 THz	2.17–2.50 eV
Yellow	570–590 nm	508–526 THz	2.10–2.17 eV
Orange	590–620 nm	484–508 THz	2.00–2.10 eV
Red	620–750 nm	400–484 THz	1.65–2.00 eV

$$E_{fh}$$

تابش جسم سیاه

(تعریف جسم سیاه، مشخصه تابشی وابسته به دما مطابق تئوری کوانتومی)

The (absolute) blackbody absorbs all energy, and reflects nothing, which is of course an idealization. A black-body at room temperature appears black, as most of the energy it radiates is infra-red and cannot be perceived by the human eye. Black-body radiation has a characteristic, continuous frequency spectrum that depends only on the body's temperature. The thermal radiation emitted by many ordinary objects can be approximated as blackbody



Classical physics can be used to derive an equation which describes the intensity as a function of frequency for a fixed temperature (Rayleigh-Jeans law). Although the Rayleigh-Jeans law works for low frequencies, it diverges for high frequencies.

تابش جسم سیاه

(قانون وین و قانون استفان - بولتزمن)

Wien's displacement law indicates that the maximum of the energy distribution is displaced within the radiation spectrum of a blackbody in case of a change in temperature.

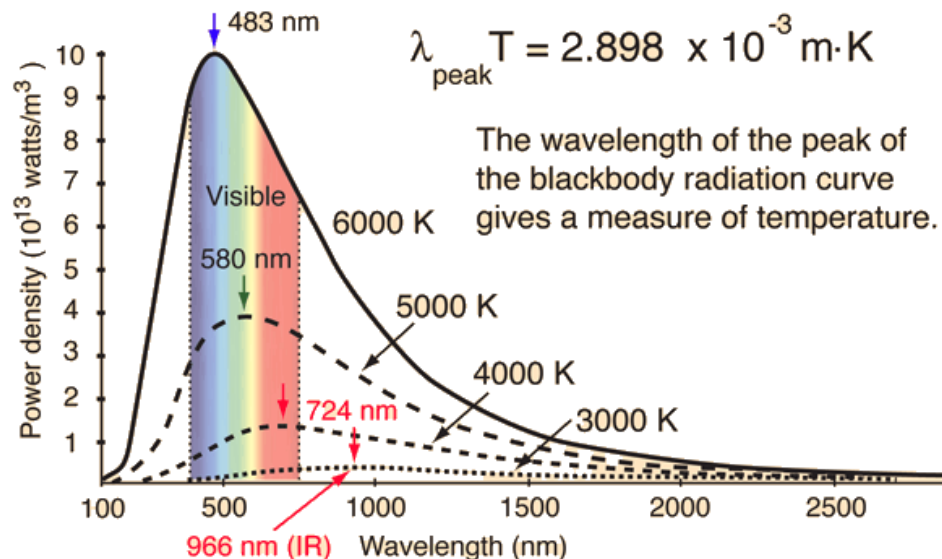
$$\lambda_{\text{max}} T = b$$

where b is called Wien's displacement constant, is equal to $2.89 \times 10^{-3} \text{ K}\cdot\text{m}$.

The **Stefan-Boltzmann law** states that the power emitted by the surface of a black body is directly proportional to the fourth power of its absolute temperature and, of course, its surface area A :

$$P = \sigma T^4 A$$

where $\sigma \approx 5.67 \cdot 10^{-8} \text{ W}/(\text{m}^2 \text{K}^4)$ is the Stefan-Boltzmann constant.



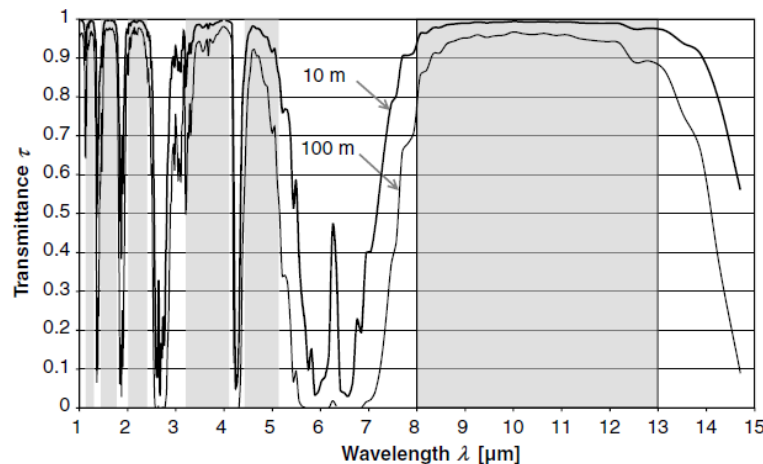
تابش جسم سیاه

سوال: بیشترین تابش بدن انسان در چه طول موجی و در چه محدوده ای از طیف الکترومغناطیسی است؟ با فرض اینکه سطح بدن یک انسان ۲ متر مربع باشد، توان تابشی چقدر است؟

Table 1.1.3 Classification of infrared radiation

Range		Wavelength λ (μm)	Wave number σ (cm^{-1})	Frequency (ν) ^a (THz)	Photon energy E^b (eV)
IR	VIS	0.38–0.78	26 316–12 821	789–384 THz	3.27–1.59
	NIR	0.78–3	12 821–3333	384–100 THz	1.59–0.41
	MIR	3–6	3333–1667	100–50 THz	0.41–0.21
	FIR	6–40	1667–250	50–7.5 THz	0.21–0.03
	UFIR	40–1000	250–10	7.5 THz–300 GHz	$0.03\text{--}1.2 \times 10^{-3}$

طیف مادون قرمز:



جذب مادون قرمز در هوا:

واحد‌های نور

(درخشش، شدت، شار نور)

Candela (cd)

Unit of luminous intensity of a light source in a specific direction. Also called *candle*.

Technically, the radiation intensity in a perpendicular direction of a surface of 1/600000 square meter of a black body at the temperature of solidification platinum under a pressure of 101,325 newtons per square meter.

Footcandle (fc or ftc)

Unit of light intensity, measured in lumens per square foot. The brightness of one candle at a distance of one foot. Approximately 10.7639 lux.

Lumen (lm)

Unit of light flow or luminous flux. The output of artificial lights can be measured in lumens.

Lux (lx)

Unit of illumination equal to one lumen per square meter. The metric equivalent of foot-candles (one lux equals 0.0929 footcandles). Also called meter-candle.



Miniature Edison Screw (MES)

Guide to source illuminations

Light source Illumination	LUX
Moonlight	0.1
60W Bulb at 1m	50
1W MES Bulb at 0.1m	100
Fluorescent Lighting	500
Bright Sunlight	30,000

برخی آثار نوری (فوتونی)

- ❑ اثر فتوالکتریک
- ❑ اثر فتوکانداکتیو
- ❑ اثر میکروبالومتر
- ❑ اثر فتوولتاییک
- ❑ اثر نور در دیود PIN
- ❑ اثر تزویج بار CCD
- ❑ اثر فتو دی الکتریک
- ❑ آثار فتولومینسانس
- ❑ اثر فولورسانس
- ❑ اثر فسفرسانس
- ❑ اثر الکترو لومینسانس

اثر فوتوالکتریک

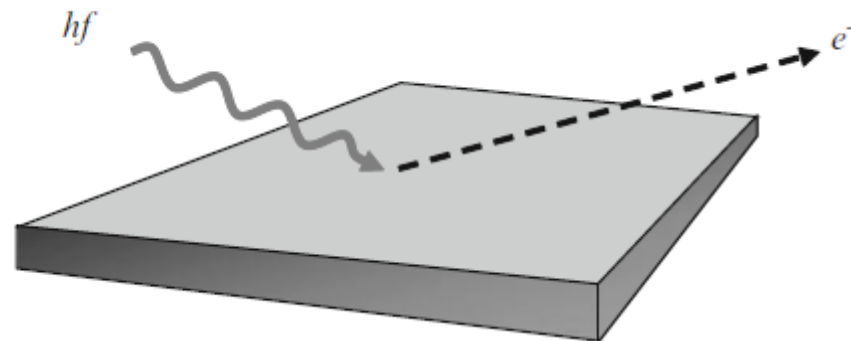
(تعریف فوتوالکتریک و تابع کار و بیان رابطه انرژی جنبشی فوتوالکتریک با تابع کار و انرژی فوتون تابیده طبق اثر فوتوالکتریک)

When a material is irradiated by photons, electrons may be ejected from it. The ejected electrons are called *photoelectrons*, and their kinetic energy, E_K , is equal to the incident photon's energy, hf , minus some threshold energy, known as the *material's work function* ϕ , which needs to be exceeded for the material to release electrons. The effect is illustrated in Fig. 3.3 and is governed by:

$$E_K = hf - \phi, \quad (3.18)$$

where h is the Planck's constant ($h = 6.625 \times 10^{-34}$ J s) and f is the photon's frequency.

Fig. 3.3 Photoelectric effect: incident of a photon and release of an electron as a result



تاریخچه اثر فتوالکتریک

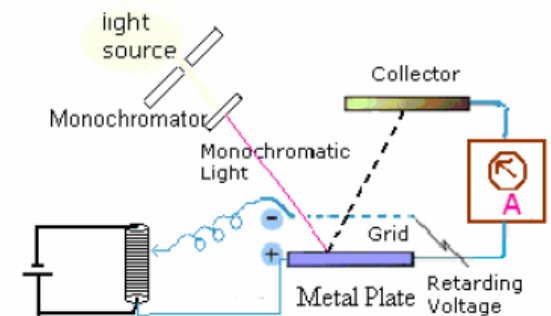


In 1887, Heinrich Hertz and his assistant Philipp Lenard observed that when light was shone on certain substances, the substances gave out cathode rays (electrons); only the number of electrons emitted, and not their energy, increased when the strength (intensity) of the incident light was increased. (Incidentally, a nephew of Heinrich Hertz, Gustav Ludwig Hertz, won the Physics Nobel Prize for 1925 along with James Franck for their discovery of the laws governing the impact of an electron upon an atom.)



Picture 4:
Philipp Lenard [13]

White light is sent through a monochromator and is then incident on a thin metal plate. The incident light is absorbed by the metal plate which then ejects electrons from it. These electrons are attracted toward a collector at which they arrive, but only if they have sufficient kinetic energy to get past a grid (a fine wire mesh) to which a retarding potential is applied, as shown in the figure. The kinetic energy of the electrons emitted from the thin metal foil could thus be gauged by varying the retarding voltage. The 'plate to collector' electron current, being measured by an ammeter. Lenard made the surprising discovery of a 'cutoff frequency'.



تاریخچه اثر فتوالکتریک

(اندازه گیری های فلیپ لنارد و بیان کوانتای انرژی برای نور توسط انیشتین)

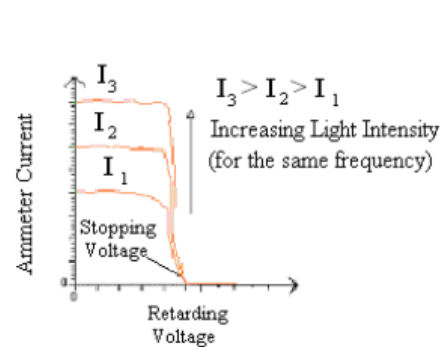


Fig. 2:

A sufficiently negative voltage on the grid prevents electrons from making it to the collector, regardless of the intensity of the incident light.

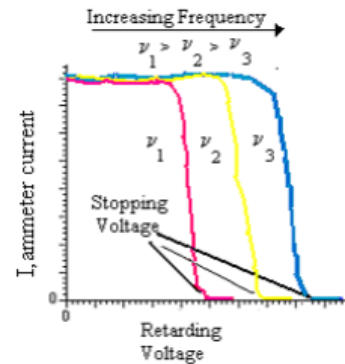
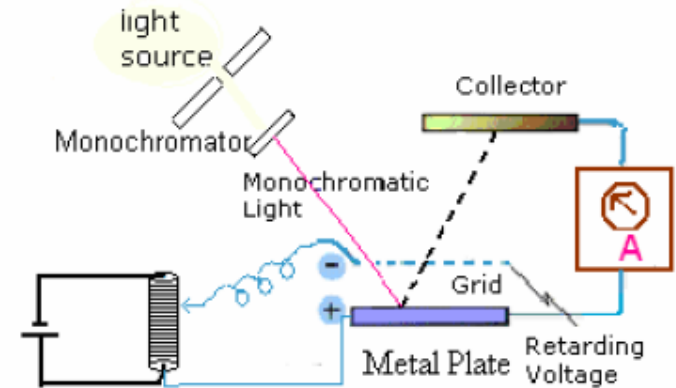


Fig. 3:

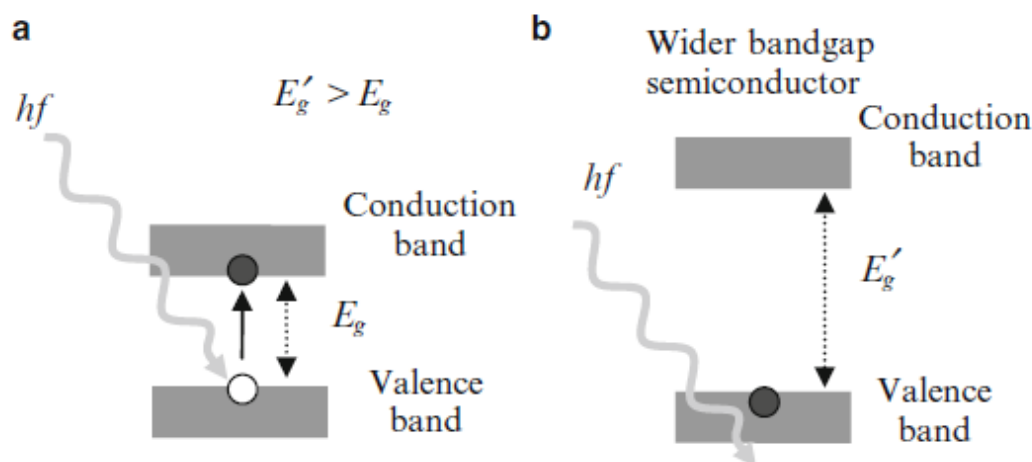
The stopping potential is higher for larger frequency, and the two have a linear relationship.



Until Einstein wrote this paper, it was assumed that electromagnetic radiation traveled as waves. Einstein considered the quantization of light into packets of energy called quanta in the context of propagation of EM energy. Wrote Einstein: "According to the assumption considered here, when a light ray starting from a point is propagated, the energy is not continuously distributed over an ever increasing volume, but it consists of a finite number of energy quanta, localized in space, which move without being divided and which can be absorbed or emitted only as a whole".

اثر فوتوکاندکتیو

Photoconductivity occurs when a beam of photons impinges a semiconducting material, causing its conductivity to change. The incident photons excite electrons from the conduction band to the valence band, which occurs if the light striking the semiconductor has sufficient energy (hf is the energy of the incident photon that has the frequency of f). Light response depends on the *bandgap* of the materials. A simple depiction of electronic band structures for two semiconductors with different bandgaps is shown in Fig. 3.4. Obviously for a wider bandgap material larger energy is required to excite an electron from the valence band to the conduction band.



حسگر فوتوکانداکتیو

(مثالهای از مواد فوتوکانداکتیو با کاربرد در طول موجهای مختلف)

The photoconductive effect is widely utilized in electromagnetic radiation sensors, and such devices are termed *photoconductors*, *light-dependent resistors (LDR)*, or *photoresistors*. *Cadmium sulfide* (CdS—bandgap of ~ 2.42 eV, which is approximately the wavelength of 512 nm in the violet region) and *cadmium selenide* (CdSe— ~ 1.73 eV, which is approximately the wavelength of 716 nm in the yellow region) are the current materials of choice for the fabrication of photoconductive devices and sensors. Very commonly to make a photoresistor, a film of these materials is deposited onto parallel electrodes (Fig. 3.5). Devices based on semiconductors such as CdS can have a wide range of resistance values, from about a few ohms, when the light has high intensity, to several mega ohms in darkness.



Fig. 3.5 Photo of a commercial LDR based on CdS

حسگر فوتوکاندکتیو

(مثال از مشخصات یک حسگر فوتوکاندکتیو)



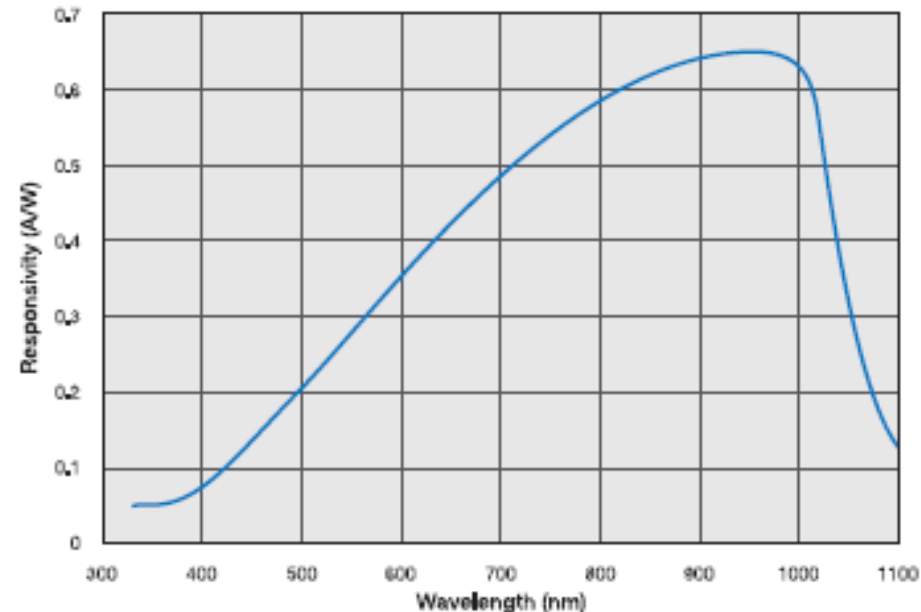
■ APPLICATIONS

- Pulse Detectors
- Optical Communications
- Bar Code Readers
- Optical Remote Control
- Medical Equipment
- High Speed Photometry

■ FEATURES

- High Speed Response
- Low Capacitance
- Low Dark Current
- Wide Dynamic Range
- High Responsivity

■ Typical Spectral Response



HAMAMATSU
PHOTON IS OUR BUSINESS



MCT photoconductive detectors

P3257 series

P4249-08

10 μm band infrared detector with high sensitivity and high-speed response

Features

- High-speed response, high sensitivity in the 10 μm band detection
- Photoconductive element that decreases its resistance by input of infrared light
- Custom devices available
Custom devices not listed in this catalog are also available with different spectral response, photosensitive area sizes and number of elements.
- Non-cooled type and thermoelectrically cooled type not requiring liquid nitrogen are also provided.
Also available are easy-to-handle infrared detector modules with preamp.

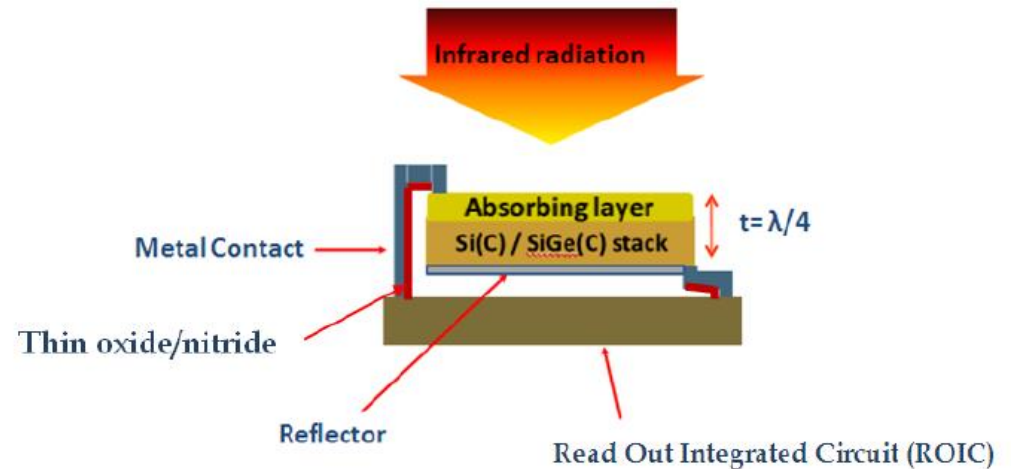
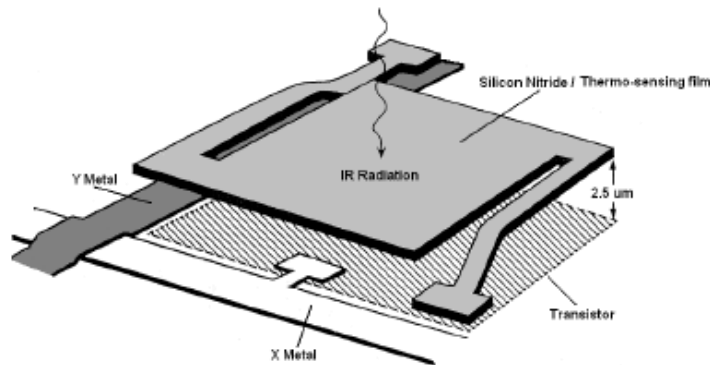
Applications

- Thermal imaging
- Remote sensing
- FTIR
- CO₂ laser detection
- Infrared spectrophotometer

Options

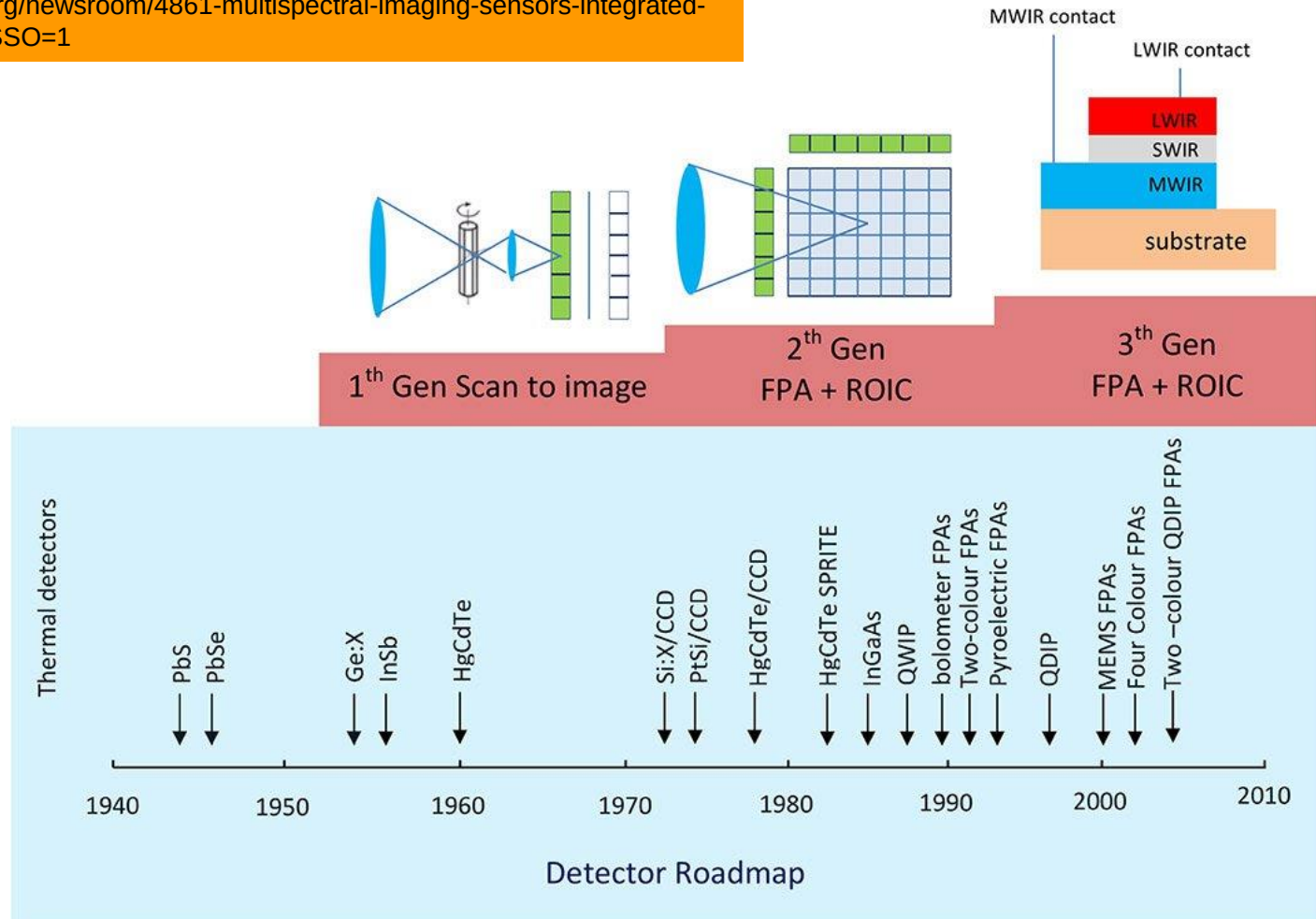
- Valve operator A3515
- Amplifiers for dewar type MCT photoconductive detector C5185-02
(The amplifie for P3257-25 is a custom-made product.)

Microbolometer is a detector for infrared radiation. When wavelengths between 7.5–14 μm strikes the detector material, heating it, and thus changing its electrical resistance.



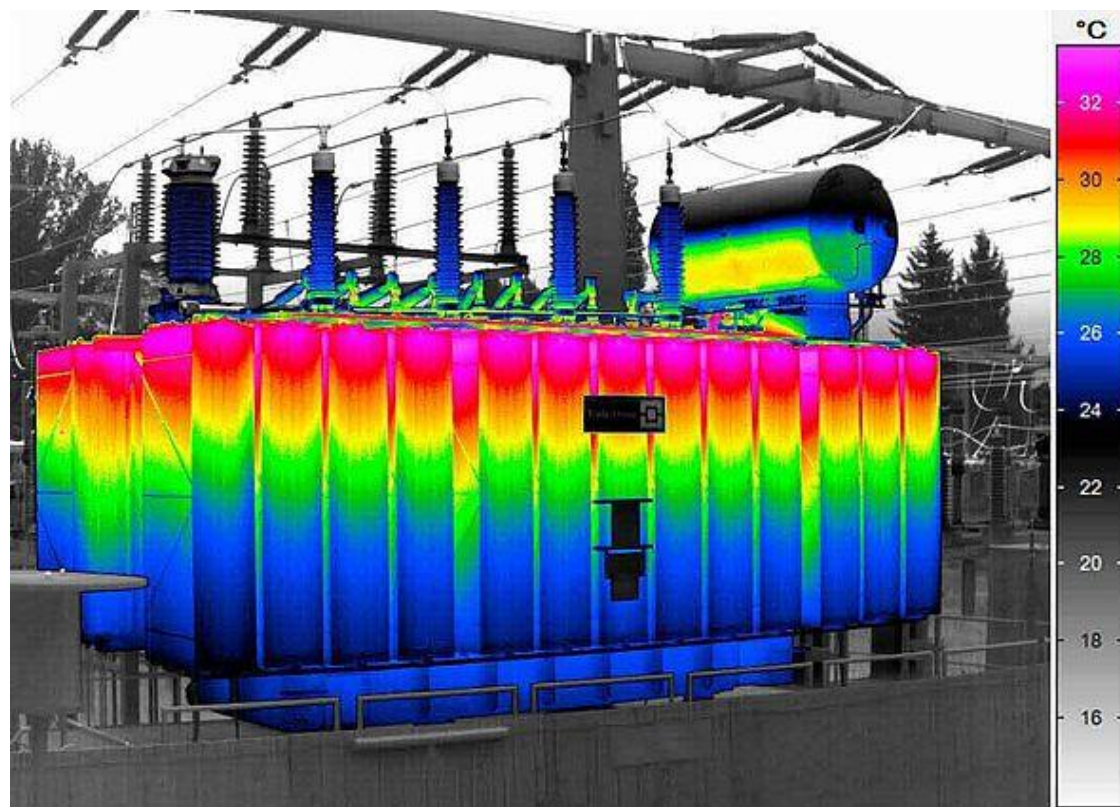
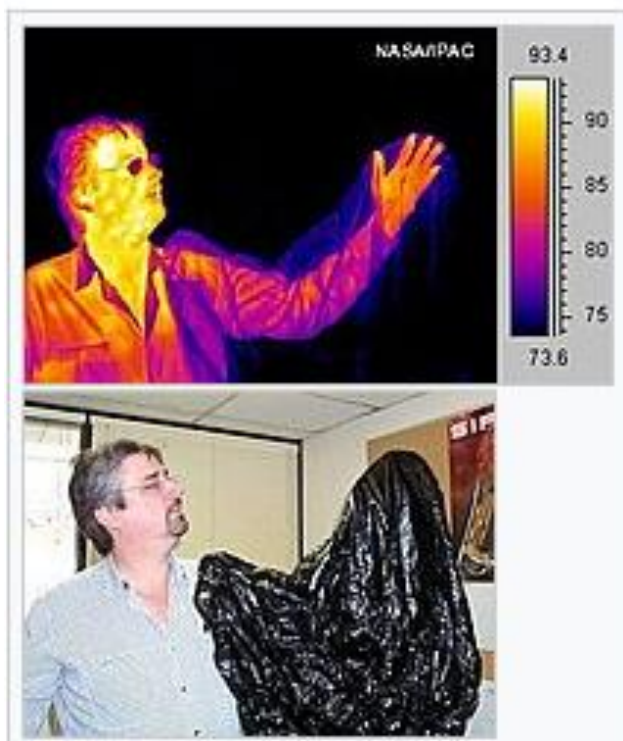
روند پیشرفت در ساخت سیستم های تصویر بردار حرارتی

<http://spie.org/newsroom/4861-multispectral-imaging-sensors-integrated-on-silicon?SSO=1>



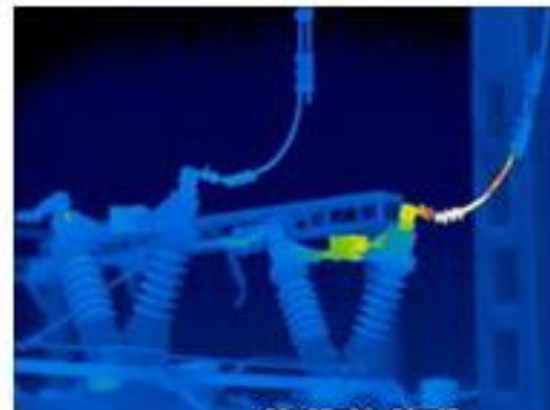
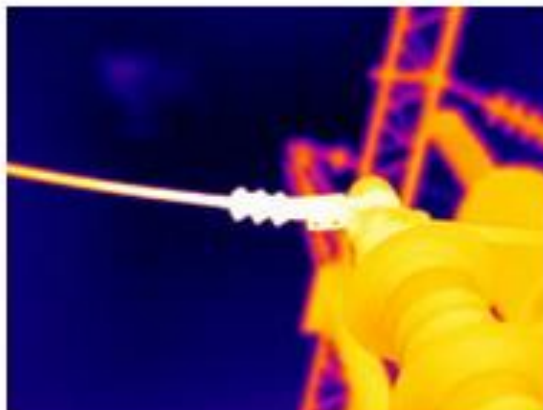
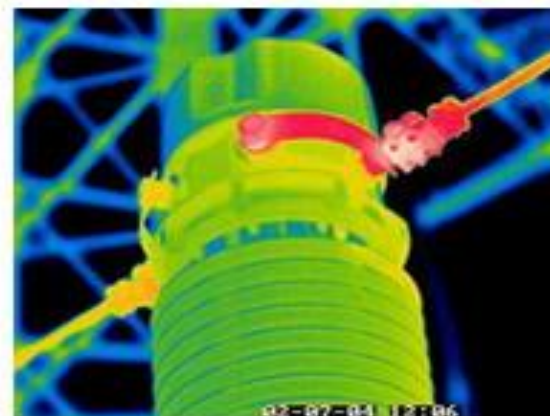
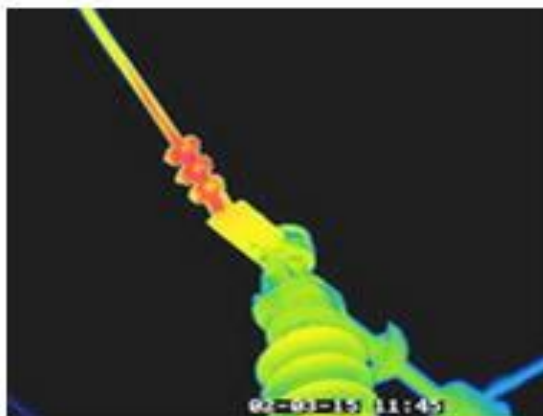
- اختراع ترانزیستور ۱۹۴۷ میلادی
- ساخت اولین مدار مجتمع ۱۹۵۸ میلادی
- ساخت اولین مدارات مجتمع خیلی فشرده VLSI سال ۱۹۷۰ میلادی

مثالهایی از تصاویر حرارتی تصویربردارهای مادون قرمز



<https://www.infratec.co.uk/thermography/industries-applications/plant-inspection/>

مثالهایی از تصاویر حرارتی تصویربردارهای مادون قرمز



SATIR Hotfind-S
Industrial Thermal Imaging Camera

<https://satir.com/article/thermal-imaging-and-electrical-industry>

دوربین حرارتی FLIR- E25 (مورد استفاده در گروه ملی فولاد)



Common Features

Detector Type and Pitch	Uncooled microbolometer, 17 μm
Thermal Sensitivity (NETD)	$<0.04^{\circ}\text{C}$ @ 30°C (86°F), 24° lens
Spectral Range	7.5 - 14.0 μm
Image Frequency	30 Hz

Image Presentation and Modes

Display	4", 640 × 480 pixel touch screen LCD with auto-rotation
Digital Camera	5 MP, 53° × 41° FOV
Color Palettes	Iron, Gray, Rainbow, Arctic, Lava, Rainbow HC
Image Modes	Infrared, visual, MSX®, Picture-in-Picture
MSX®	Embosses visual details on full resolution thermal image

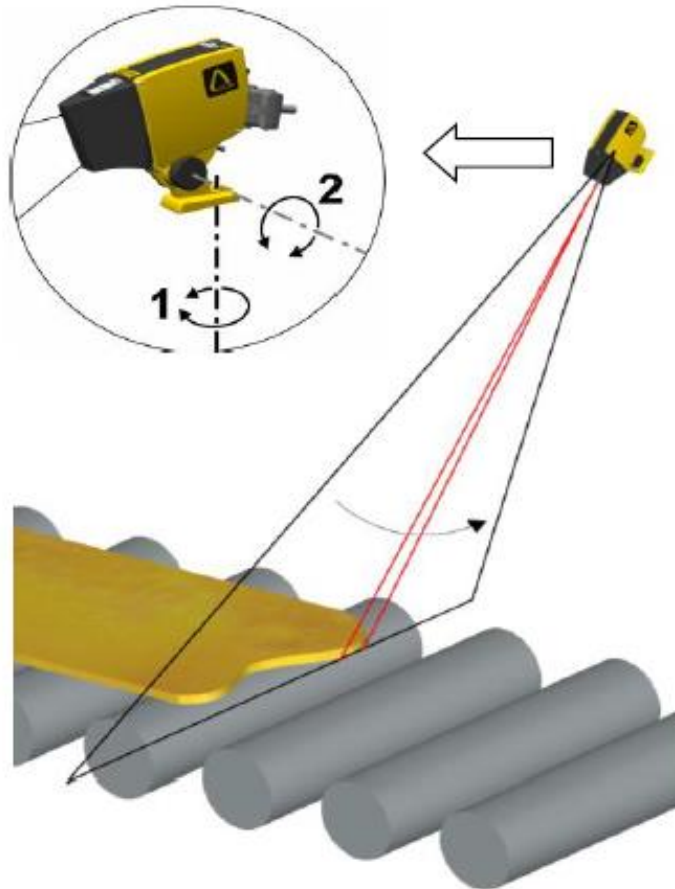
Measurement and Analysis

Accuracy	$\pm 2^{\circ}\text{C}$ ($\pm 3.6^{\circ}\text{F}$) or $\pm 2\%$ of reading for ambient temperature 15°C to 35°C (59°F to 95°F) and object temperature above 0°C (32°F)
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حسگر تشخیص حضور فلز داغ (مورد استفاده در گروه ملی فولاد)

Infrared Scanning for high detection accuracy:

The Rota-Sonde DC4500 scans the infrared radiation emitted by hot products within a narrow lateral field with different angles of 10° , 30° or 50° depending on the model.





حسگر تشخیص حضور فلز داغ

Optics

Model	DC451●-●●	DC453●-●●	DC455●-LP
Scanning angle (1)	10°	30°	50°
Response time	1 ms	2 ms	4 ms
Sensitivity with 2 mm shutter (2)	High sensitivity range: 220 °C (430 °F) Normal sensitivity range: 400 °C (750 °F)		
Sensitivity shutter removed (2)	High sensitivity range: 180 °C (360 °F)		
Spectrum sensitivity	1 to 2.6 μm with maximum sensitivity at 2.2 μm		

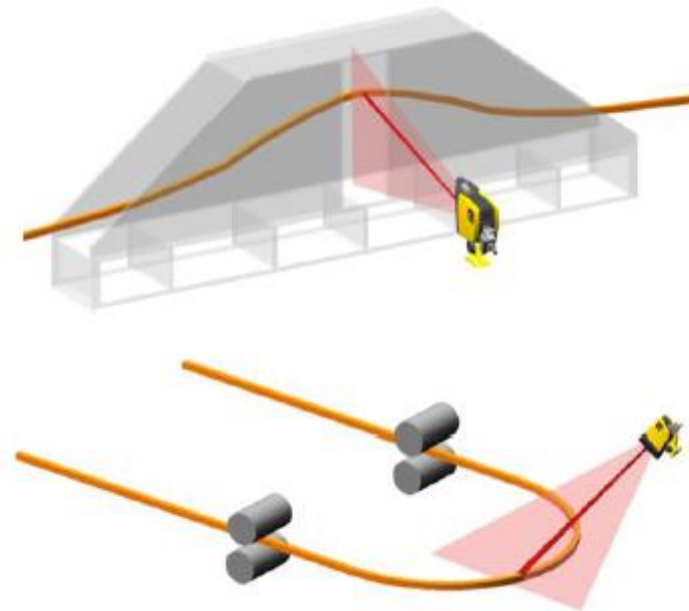
Distance between sensor and product

	Low temperature application	Standard application
Wire	0.1 to 0.5 m	0.3 to 2 m
Bar	0.3 to 1 m	0.5 to 6 m
Billet - Bloom	0.5 to 2 m	1 to 10 m
Slab - Strip - Plate	2 to 4 m	3 to 20 m

حسگر تشخیص مکان آهن گداخته (مورد استفاده در گروه ملی فولاد)

Rota-Sonde

TS5500

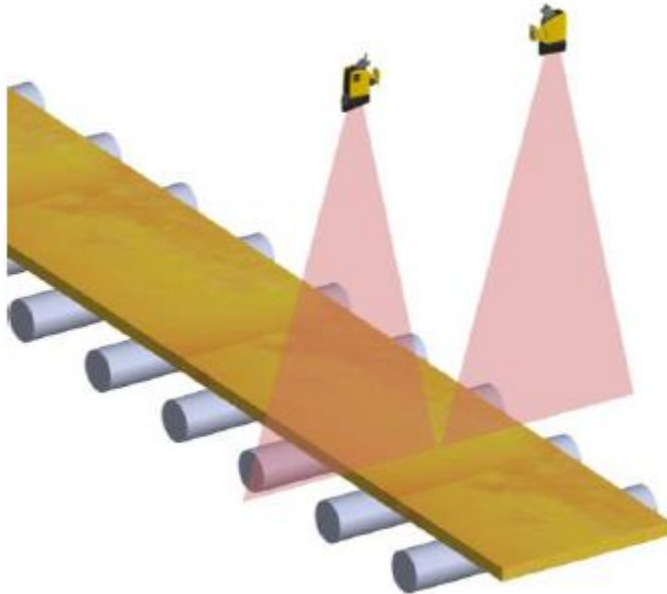


Infrared Loop Scanner

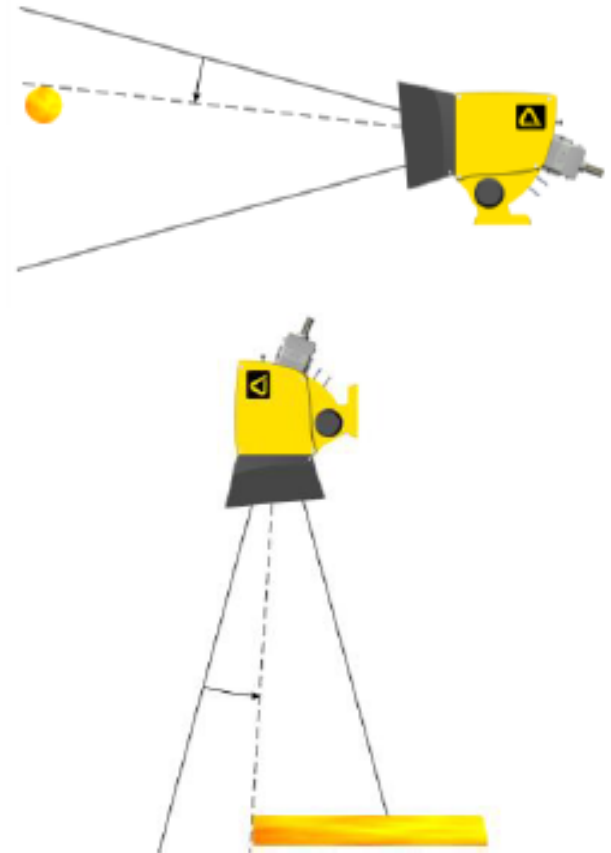
The Rota-Sonde TS5500 measures the position of a hot product (steel, copper, brass, other alloys) within its field of view and provides an output which is proportional to the angular position of the product within that field.

حسگر تشخیص مکان آهن گداخته

The **Rota-Sonde TS5500** is sensitive to infrared radiation and is optimized for detection even at low temperature and in the presence of steam. The sensor detects the first edge of the product within the scanned field, i.e. the first transition "no radiation / presence of radiation", direction of scanning being from the top to the bottom of sensor.



Hot strip, plate, slab centring - Slab edges control for torch machine
Edge position feedback allows centreline & width measurement.



<https://www.deltasensor.eu/en/Products/Infrared-Loop-Scanners/Rota-Sonde-TS5500>

اثر فوتوولتائیک

In the photovoltaic effect, a voltage is induced by absorbed photons at a junction of two dissimilar materials (it is also called a *heterojunction*). The absorbed photons produce free charge carriers (electrons and holes) as can be seen in Fig. 3.6.

A typical photovoltaic device is seen in Fig. 3.6. They generally consist of a large area semiconductor p–n junction or diode. A photon impinging on the junction is absorbed, if its energy is greater than or equal to the semiconductor's bandgap energy. This can cause a valance band electron to be excited into the conduction band, leaving behind a hole, and thus creating a mobile electron–hole pair. If the electron–hole pair is located within the depletion region of the p–n junction, then the existing electric field will either sweep the electron to the n-type side or the hole to the p-type side. As a result, a current is generated as below:

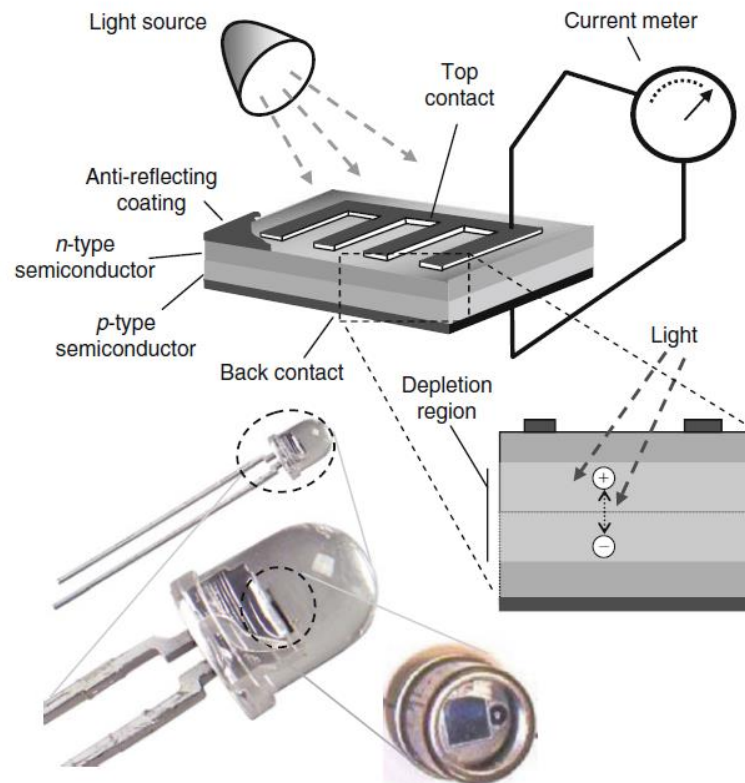
$$I = I_S [e^{\frac{qV}{kT}} - 1], \quad (3.19)$$

where q is the electron charge (1.602×10^{-19} C), k is the Boltzmann's constant (1.38×10^{-23} J/K⁻¹), and T is the temperature of the p–n junction in Kelvin.

حسگر فوتوولتائیک

(مواد به کار رفته)

Photovoltaic cells and sensors are commonly made from materials that absorb photons in the infrared, visible, and UV ranges: materials such as silicon (wavelengths between 190 and 1,100 nm), germanium (800–1,700 nm), indium gallium arsenide (800–2,600 nm), and lead sulfide (1,000–3,500 nm) are generally used.

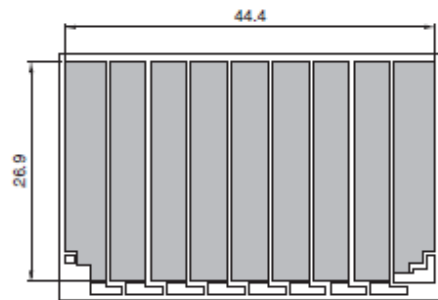


سلول فوتوولتائیک

(مثالی از یک نمونه تجاری)

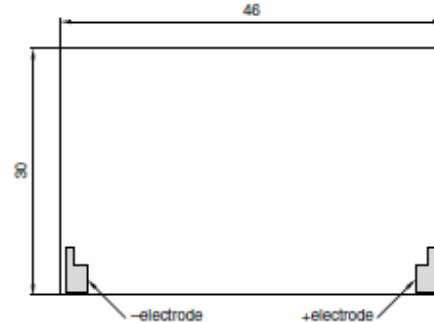
DESIGN EXAMPLES BCS4630B9 SHAPES AND DIMENSIONS

Photoactive side
Hatching areas: Photoactive area



Dimensions in mm

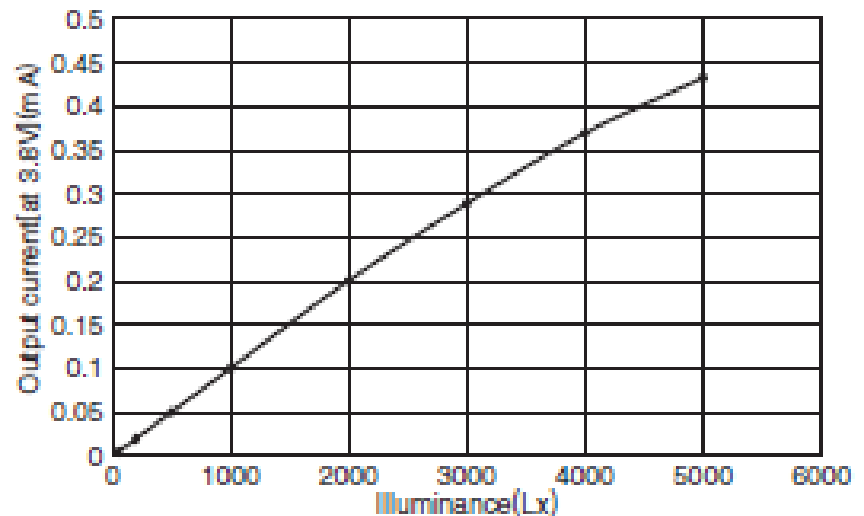
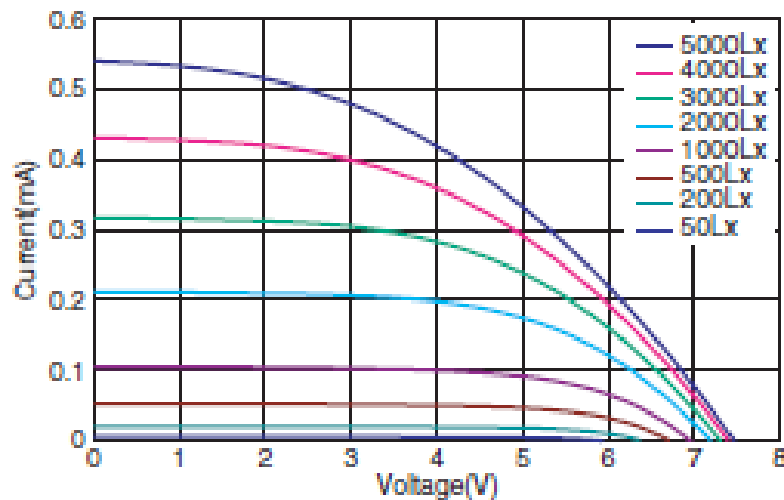
Electrode side
Hatching areas: Electrodes



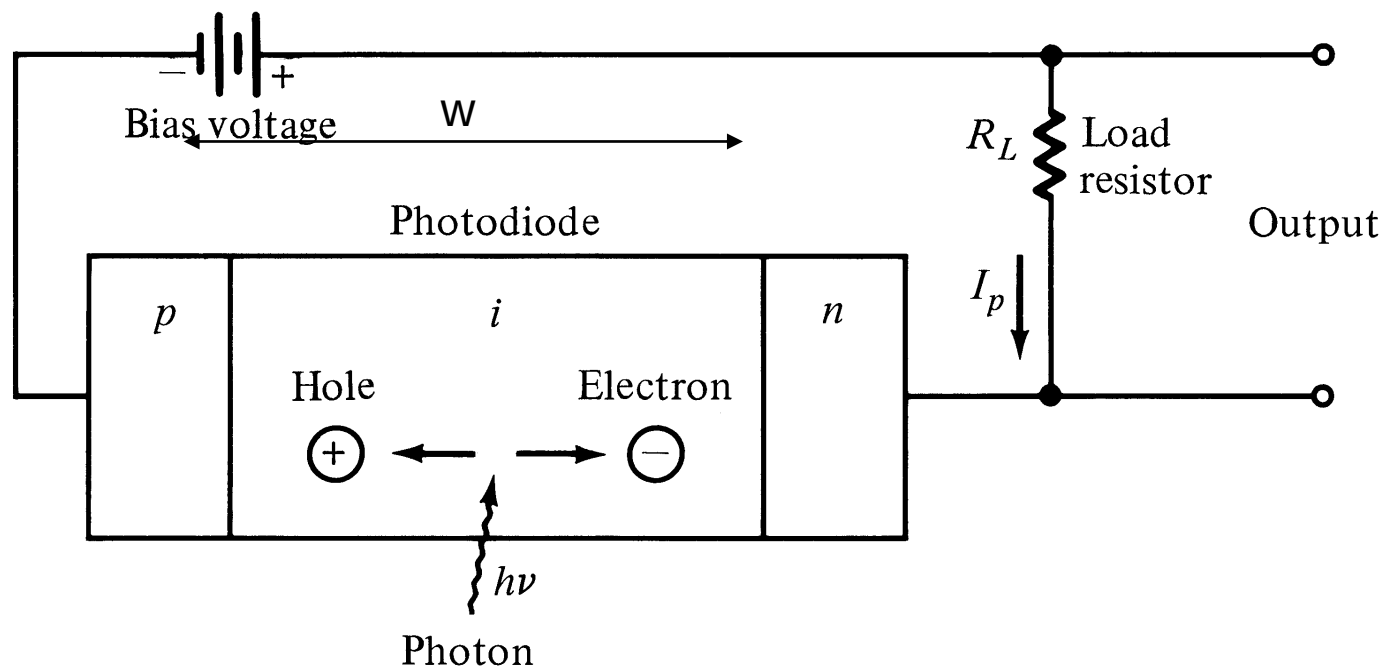
Dimensions in mm

APPLICATIONS

- Wristwatches
- Multifunctional IC cards with display
- Energy harvesting elements
- Digital books
- Charging and power supply applications for other electronic devices



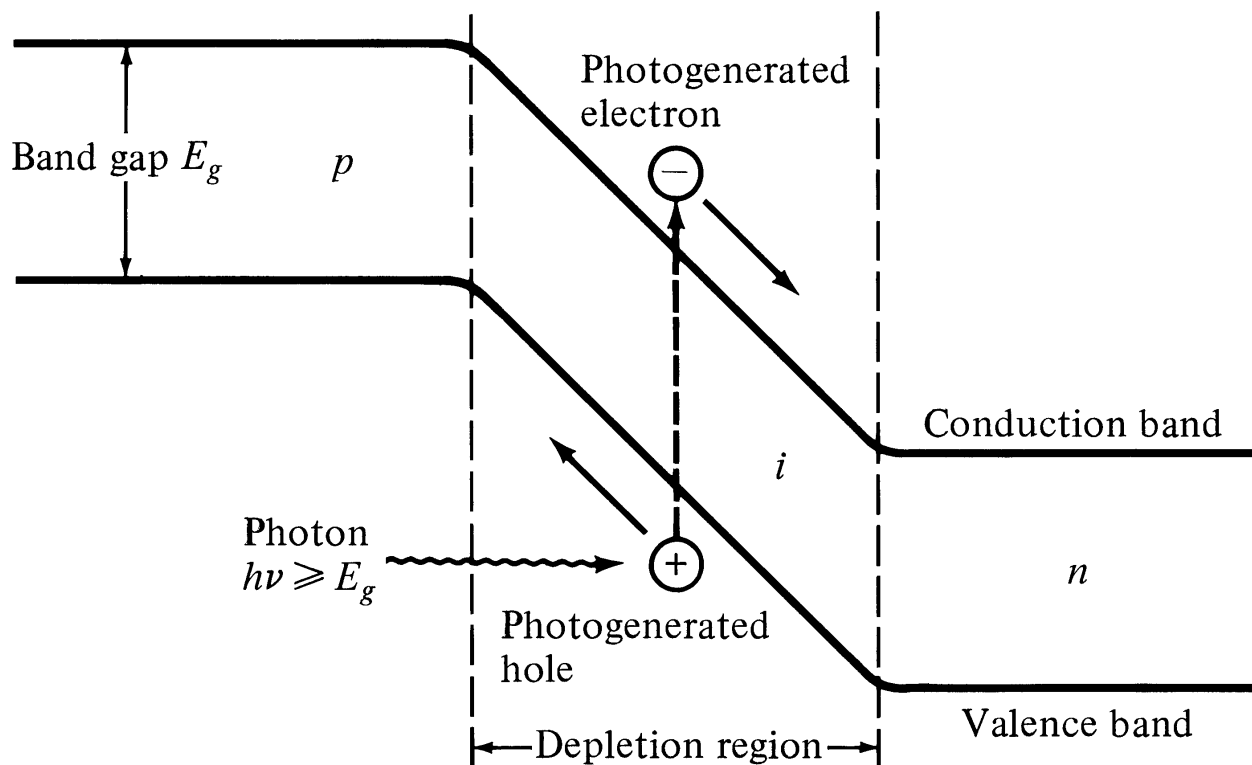
pin Photodetector:



The high electric field present in the depletion region causes photo-generated carriers to separate and be collected across the reverse –biased junction. This give rise to a current flow in an external circuit, known as **photocurrent**.

حسگر PIN:

نمایش نوارهای انرژی با اعمال بایاس و ساز و کار حسگری در PIN:

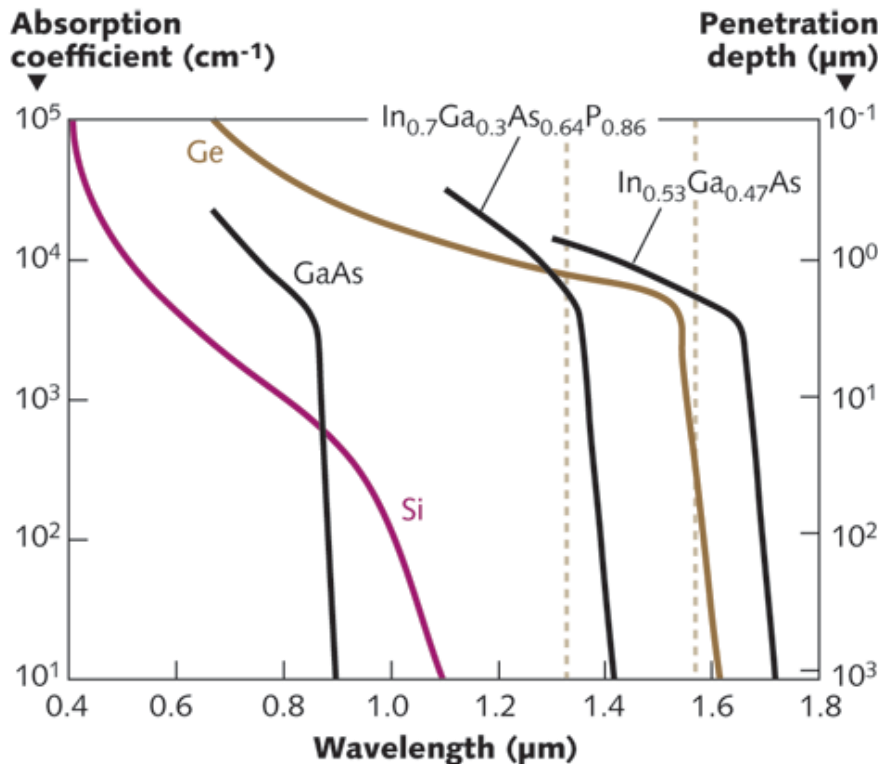


جذب نور در ناحیه تخلیه:

- Absorption strongly depends on wavelength. The upper wavelength cutoff for any semiconductor can be determined by its energy gap as follows:

$$\lambda_c = \frac{hC}{E_g} \quad \lambda_c (\mu\text{m}) = \frac{1.24}{E_g (\text{eV})}$$

وابستگی ضریب جذب نوری مواد مختلف به طول موج:



کمیت‌های مقایسه‌ای حسگرهای نوری:

- محدوده پاسخدهی
- طول موج قطع (λ_c)
- جریان تاریک (جریان نور نخورده)
- پاسخدهی (Responsivity)
- بازده کوانتومی (Quantum Efficiency)
- توان معادل نویز (NEP) Noise Equivalent Power
- گیرندگی (Detectivity)

پاسخدهی و بازده کوانتومی:

□ Responsivity:

$$R = \frac{I_P}{P_0} \quad [A/W]$$

□ Quantum Efficiency:

$$\eta = \frac{\text{\# of electron-hole photogenerated pairs}}{\text{\# of incident photons}}$$

□ Responsivity Vs. Quantum efficiency:

$$\eta = \frac{I_P / q}{P_0 / h\nu} = R \frac{h\nu}{q} \quad \rightarrow \quad R = \eta \frac{q}{h\nu}$$

NEP و گیرندگی:

□ Noise Equivalent Power:

parameter defined as the radiant power incident on the detector that produces a signal equal to the root mean square (rms) detector noise.

□ Detectivity:

$$D = \frac{1}{NEP}$$

□ Normalize Detectivity (D^*):

sorbing (active) area, A_d , and the signal bandwidth, B , one can define specific (or normalized) detectivity, D^* , as^[4]

$$D^* = \frac{\sqrt{A_d B}}{NEP}$$

تشکر از توجه شما

